

Find and Fix: Separable Equations and Initial Conditions

Answer Key

AP Calculus AB/BC

Quick Answers

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| 1. the corrected integration = $\frac{x^2}{2}$, — | 5. the corrected $P(15) = 4919.21$, — |
| 2. the corrected $y(1) = 29.556$, — | 6. the corrected $y(1) = 3.946$, — |
| 3. the corrected $y(10) = 51.723$, — | 7. the corrected y at $\pi/2 = 3.162$, — |
| 4. the corrected $y(2) = 5.0$, — | 8. the corrected $y(2) = 3.899$, — |
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What is verified

8 of 16 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-7 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

2. If they got 1613.715: integrated $6x^2$ as $6x^3$ instead of $2x^3$
 3. If they got 13.389: added the constant after exponentiating instead of multiplying by e to the C
 4. If they got 4.123: integrated $y \, dy$ as y^2 instead of y^2 over 2
 5. If they got 2002.46: added the starting balance instead of multiplying by it
 7. If they got 2.0: used the starting value 3 as the constant instead of its square 9
 8. If they got 5.325: read the constant off the squared side without doubling the halved equation correctly
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1. A student separates $\frac{dy}{dx} = \frac{x}{y}$ as $y \, dy = x \, dx$, then integrates to get $y^2 = x^2 + C$. (a) Give the correct integrated form. (b) Name the error.

Solution (a): The separation is fine. Integrating, $\int y \, dy = \frac{1}{2}y^2$ and $\int x \, dx = \frac{1}{2}x^2$.

$\frac{1}{2}y^2 = \frac{1}{2}x^2 + C$

Solution (b) — instructor-judged. Both antiderivatives carry a factor of one half, and the student dropped both. Writing $y^2 = x^2 + C$ happens to be a correct *rearrangement*

of $\frac{1}{2}y^2 = \frac{1}{2}x^2 + C$ only if the two halves are cancelled deliberately and the constant is renamed — which is a separate algebra step, not part of integrating. Full credit names the missing one half on both sides.

- 2.** To solve $\frac{dy}{dx} = 6x^2y$ with $y(0) = 4$, a student writes $\ln|y| = 6x^3 + C$ and reports $y(1) = 4e^6 \approx 1613.715$. (a) Correct $y(1)$. (b) Name the error.

Solution (a): Separating, $\frac{dy}{y} = 6x^2 dx$, and $\int 6x^2 dx = 2x^3$. So $\ln|y| = 2x^3 + C$ and $y = Ae^{2x^3}$; from $y(0) = 4$, $A = 4$, giving $y = 4e^{2x^3}$. Then $y(1) = 4e^2 = 29.5562\dots$

$y(1) \approx 29.556$

Solution (b) — instructor-judged. The power was raised but never divided by: the antiderivative of $6x^2$ is $\frac{6}{3}x^3 = 2x^3$, not $6x^3$. The reported exponent 6 is the visible symptom. Full credit states the power rule for antiderivatives and points at that exponent.

- 3.** For $\frac{dy}{dx} = \frac{y}{5}$ with $y(0) = 7$, a student integrates to $\ln|y| = \frac{x}{5} + C$, then writes $y = e^{x/5} + C$, gets $C = 6$, and reports $y(10) \approx 13.389$. (a) Correct $y(10)$. (b) Name the error.

Solution (a): Exponentiating $\ln|y| = \frac{x}{5} + C$ gives $y = e^{x/5} e^C = Ae^{x/5}$. From $y(0) = 7$, $A = 7$, so $y = 7e^{x/5}$ and $y(10) = 7e^2 = 51.7234\dots$

$y(10) \approx 51.723$

Solution (b) — instructor-judged. Exponentiating turns an *added* constant into a *multiplied* one, because $e^{u+C} = e^u e^C$. The student carried the $+C$ through the exponential unchanged. Full credit states that rule and uses it to get $A = 7$; half credit for noticing that $y = e^{x/5} + 6$ does not satisfy the original equation.

- 4.** A student solves $\frac{dy}{dx} = \frac{4x}{y}$, $y(0) = 3$, $y > 0$, writing $y dy = 4x dx$, then $y^2 = 2x^2 + C$, $C = 9$, $y(2) \approx 4.123$. (a) Correct $y(2)$. (b) Name the error.

Solution (a): $\int y dy = \frac{1}{2}y^2$, so the implicit solution is $\frac{1}{2}y^2 = 2x^2 + C$, i.e. $y^2 = 4x^2 + K$. From $y(0) = 3$, $K = 9$, so $y = \sqrt{4x^2 + 9}$ and $y(2) = \sqrt{16 + 9} = \sqrt{25}$.

$y(2) = 5$

Solution (b) — instructor-judged. The separation is right; the y -side integration is not. The antiderivative of y is $\frac{1}{2}y^2$, and dropping the half changed the coefficient of x^2 from 4 to 2 when the equation was doubled. The value $C = 9$ is then correct arithmetic on a wrong equation, so it is a consequence of the one error, not a second one. Full credit says both halves of that.

- 5.** A balance P in dollars satisfies $\frac{dP}{dt} = 0.06P$, $P(0) = 2000$, t in years. A student writes $P = e^{0.06t} + 2000$ and reports $P(15) \approx 2002.46$. (a) Correct $P(15)$. (b) Name the error and the sanity check.

Solution (a): $\frac{dP}{P} = 0.06 dt$ gives $\ln P = 0.06t + C$, so $P = Ae^{0.06t}$ with $A = P(0) = 2000$.

Then $P(15) = 2000e^{0.9} = 4919.2062\dots$

$$P(15) \approx 4919.21$$

Solution (b) — instructor-judged. Same exponentiation slip as problem 3: the starting balance must multiply the exponential, not be added to it. The sanity check is the story — the student's answer has fifteen years of six percent growth adding about two dollars and forty-six cents to a two thousand dollar balance, which no exponential model would produce. Full credit gives the rule and the plausibility check.

6. A student says $\frac{dy}{dx} = x + xy$ cannot be separated because the right side is a sum. (a) Separate it, solve with $y(0) = 2$, give $y(1)$. (b) Explain the test.

Solution (a): Factor: $x + xy = x(1 + y)$. Then $\frac{dy}{1 + y} = x dx$, so $\ln |1 + y| = \frac{x^2}{2} + C$ and $1 + y = Ae^{x^2/2}$. From $y(0) = 2$, $A = 3$, so $y = 3e^{x^2/2} - 1$. Then $y(1) = 3e^{1/2} - 1 = 3.9461\dots$

$$y(1) \approx 3.946$$

Solution (b) — instructor-judged. A sum does not block separation; an unfactorable sum does. The test: factor the right side and ask whether it becomes (something in x) times (something in y). Here it does. Compare $\frac{dy}{dx} = x + y$, which does not factor that way and is genuinely not separable. Full credit states the factoring test; half credit for asserting separability with no test.

7. Two students solve $\frac{dy}{dx} = \frac{\cos x}{2y}$, $y(0) = 3$, $y > 0$. One writes $y^2 = \sin x + 9$, the other $y^2 = \sin x + 3$ with $y(\frac{\pi}{2}) = 2$. (a) Correct value at $x = \frac{\pi}{2}$. (b) Which is wrong, and why?

Solution (a): $2y dy = \cos x dx$ integrates to $y^2 = \sin x + C$. At $x = 0$, $y = 3$: $9 = 0 + C$, so $C = 9$ and $y^2 = \sin x + 9$. At $x = \frac{\pi}{2}$, $y = \sqrt{1 + 9} = \sqrt{10} = 3.1622\dots$

$$y(\pi/2) \approx 3.162$$

Solution (b) — instructor-judged. The second solution is wrong: it substituted the starting value into the equation as if the left side were y rather than y^2 , so it recorded $C = 3$ instead of $C = 3^2 = 9$. Full credit identifies that solution and says the constant is found from y^2 .

8. Student work on $\frac{dy}{dx} = \frac{2x}{y+1}$, $y(0) = 3$: Line 1 $(y+1)dy = 2x dx$; Line 2 $\frac{(y+1)^2}{2} = x^2 + C$; Line 3 $C = 16$; Line 4 $y = -1 + \sqrt{2x^2 + 32}$, $y(2) \approx 5.325$. (a) Correct $y(2)$. (b) Which line is wrong?

Solution (a): Lines 1 and 2 are correct. Substituting $(0, 3)$ into Line 2: $\frac{(3+1)^2}{2} = 0 + C$, so $C = \frac{16}{2} = 8$. Then $(y+1)^2 = 2x^2 + 16$ and $y = -1 + \sqrt{2x^2 + 16}$. At $x = 2$, $y = -1 + \sqrt{24} = 3.8989\dots$

$$y(2) \approx 3.899$$

Solution (b) — instructor-judged. Line 3 is the wrong one. The student read C off the numerator $(3+1)^2 = 16$ and forgot that Line 2 divides it by two, so $C = 8$, not 16. Lines

1, 2 and 4 are each correct given what precedes them — Line 4 is a faithful rearrangement of a wrong constant. Full credit names Line 3, gives $C = 8$, and says the last line follows correctly from the corrected constant.